JNTUH USED 28-04-18AM Set-1

R13

Code No: 118BJ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, April - 2018
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) Give few examples of embedded systems. [2]
b) Give example bit address and byte address of 8051 RAM? [3]
c) Explain ADC and SUBB instructions of 8051 microcontroller. [2]
d) Explain any three logical instructions with suitable example. [3]
e) List two reasons why stop bits are used in asynchronous communication. [2]
f) Explain about Task and Task States. [3]
g) What is the function of ROM Emulator? [2]
h) Explain about Cross-compiler. [3]
i) What is a CAN Bus? [2]
j) What is meant by Networked embedded systems. [3]

PART - B

(50 Marks)

2.a) What is an embedded computer system? Explain the characteristics of embedded Computing applications.
b) Given an overview of 8051 input/output ports. [5+5]

OR

3.a) Draw the format of TMOD and TCON registers in 8051 and explain the timer/counter operation with suitable diagrams.
b) Calculate the amount of time delay generated by timer in mode 1, if it is initially set to FFF2H and the clock frequency is 11.0592 MHz. [5+5]

4.a) Explain the different addressing modes of 8051 with suitable examples.
b) Explain the instructions used to accessing the external RAM and the on-chip ROM of the 8051. [5+5]

OR

5.a) Explain the multiply and divide instructions in 8051. Write a program for addition of 5 BCD numbers stored at RAM locations 40H to 44H.
b) What are the sequence of events involved in CALL instruction. Distinguish between LCALL and ACALL instructions. [5+5]

6.a) Discuss about any two modes of serial data communication in 8051.
b) Write short notes on A/D and D/A conversion in 8051. [5+5]

OR

7.a) What is a semaphore? Explain with an example how semaphores solve the Shared-data problem.
b) Explain the features of the following methods of inter task communication.
i) Message queues, ii) Mail boxes, iii) Pipes. [5+5]

8.a) Discuss the hard real-time scheduling considerations in hard real-time systems.
b) Explain the need for memory management. Explain about memory management in an embedded system. [5+5]

OR

9.a) Describe the features of various Contemporary Real Time Operating Systems.
b) Write short notes on debugging techniques. [5+5]

10.a) Draw and explain the working of basic programming model of the ARM and SHARC processors.
b) Explain the structure and operation of the I^2C bus with the help of a state transition graph and bus timing diagram. [5+5]

OR

11.a) Explain about memory organization and Instruction level parallelism in advanced architectures.
b) Give a note on Internet-enabled systems. [5+5]

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